Age-Related Differences in Neural Responses to Trust are Associated with Risk for Financial Exploitation

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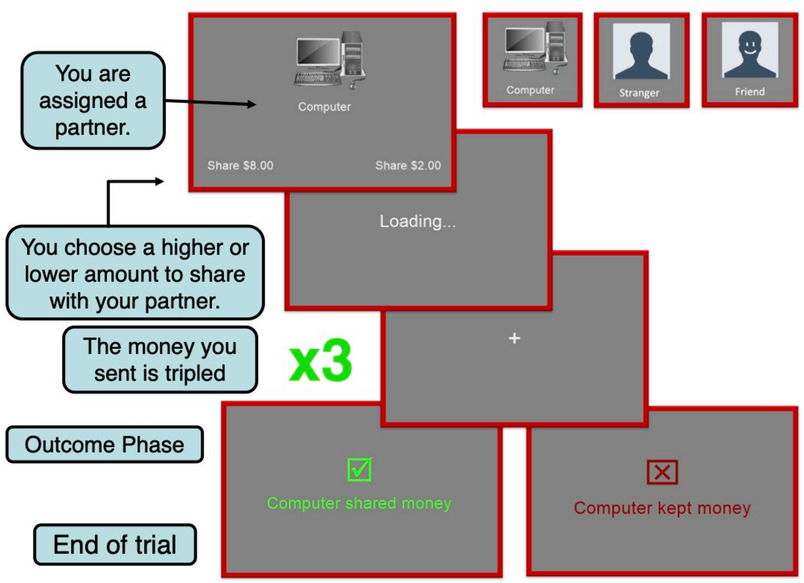
1. Introduction

- Social relationships influence our behaviors and modulate our neural responses to reward (Fareri et al., 2015).
- The effect of relationships on behavior and reward activation change as we age (Fareri et al., 2022; Cassidy et al., 2020).
- It remains unclear whether age-related differences in responses to trust are associated with maladaptive outcomes such as financial exploitation.

We investigated whether responses to trust change as we age, and if those changes could be related to financial exploitation.

2. Methods

Participants (N = 123) completed a trust task while undergoing fMRI with three partners: a computer, a stranger, and their chosen friend.



We used the Older Adults Financial Exploitation Measure (OAFEM) to measure risk for financial exploitation (Conrad et al., 2010).

Whole brain results were thresholded at Z > 3.1 and Family Wise Error corrected at p > .05.

fMRI data were preprocessed with fMRIprep and statistical analyses were performed using FSL.

Scan the QR code to access the poster reprint, pre-registration, code, and open dataset!

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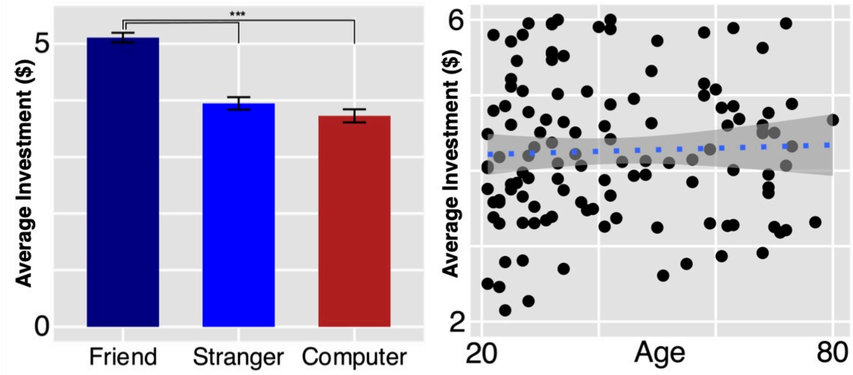
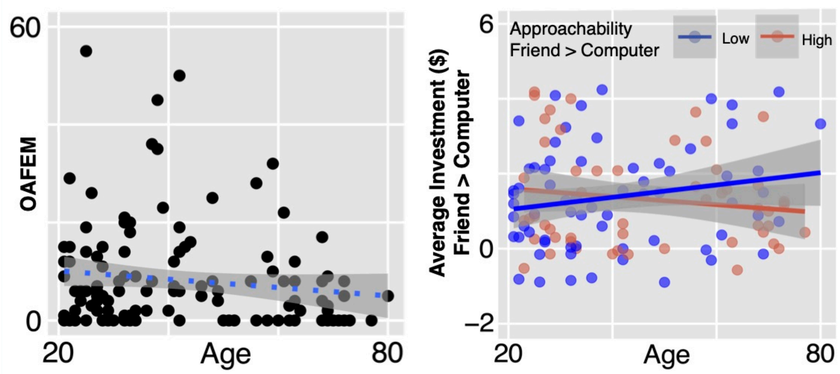
3. Age-Related Effects in Partner Perception



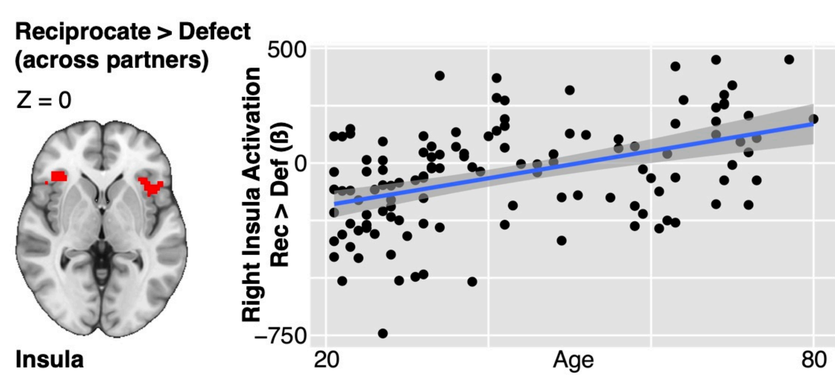
Participants invested more in their friend than the stranger or computer partners. There is no relationship between age and average investment.



The average amount that a participant invested in their friend relative to the computer was moderated by approachability ratings ($p < .05$).

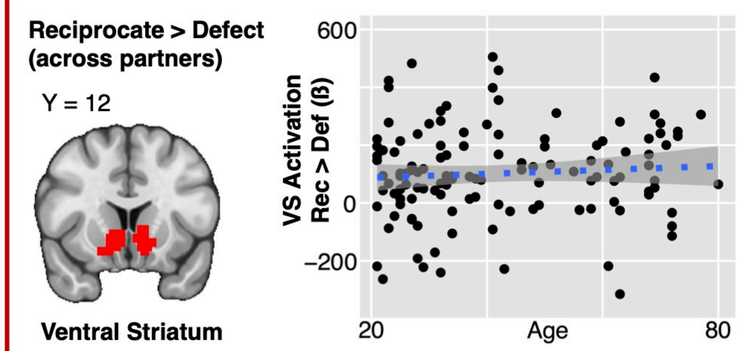
4. Age-Related Differences in Insula Response

We first investigated age-related differences in the response to partners reciprocating relative to defecting through a set of whole-brain analyses.

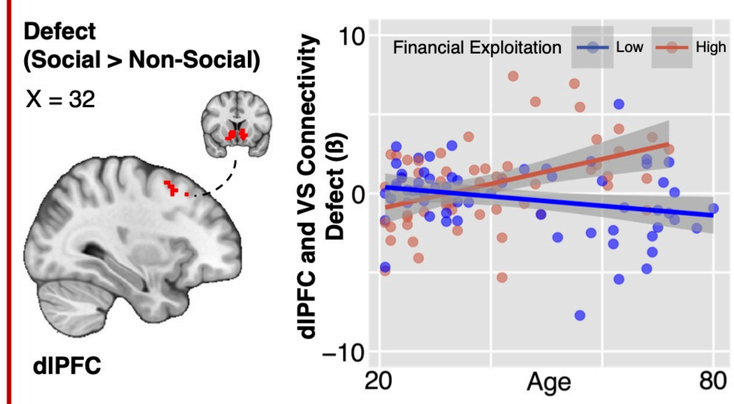


There was a positive association between age and insula response when the partner reciprocated relative to defecting.

5. Age-Related Differences in Corticostriatal Responses to Reciprocated Trust



Though there was no relationship between age and ventral striatum (VS) response to reciprocate relative to defect, we performed psychophysiological interaction (PPI) analyses using the VS as a seed region.



Risk for financial exploitation moderated VS and dorsolateral prefrontal cortex (dlPFC) connectivity when the partner defected in social conditions relative to non-social conditions.

6. Conclusions and Future Directions

- Our preliminary results indicated that neural responses to trust are associated with age and risk for financial exploitation.
- Additionally, risk for financial exploitation in older adults moderated the connectivity between the VS and dlPFC when presented with a defect outcome during the social relative to non-social conditions.
- Future analyses will examine how neural and behavioral responses to trust are associated with risk for financial exploitation.